

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Emergency Response/ Disaster Management Plan

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R01	10.08.22	Issued for Review
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Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-THS-PLN-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

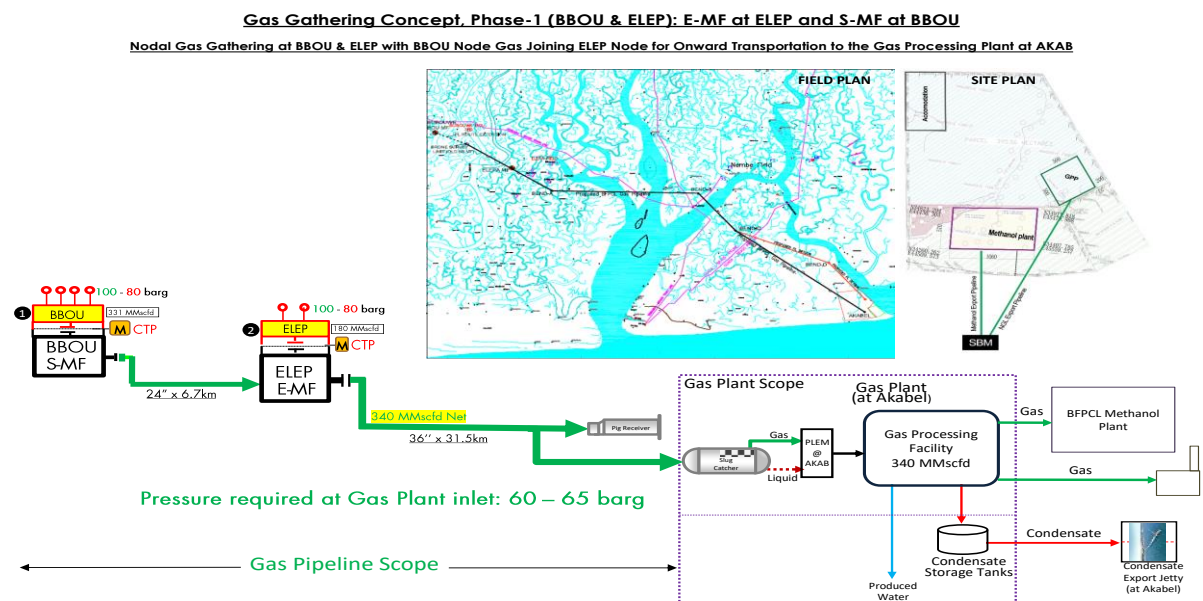


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This procedure defines the basic course of actions in emergency situations during the Execution and Operations phases of the Gas Gathering Pipelines and Associated Infrastructure Project.

The FEED shall fully comply with all specified relevant contractual requirements.

1.3. Scope

The procedure is applicable to COMPANY, Contractors, Sub-Contractors, Vendors and all personnel performing work on the facilities.

Strict adherence to this procedure will be expected during emergencies.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Emergency Response

3.1. Emergency Rescue Team

The rescue team in an emergency shall include the following:

- Site/Operations Supervisor-Leader
- Site/Operations Safety Officer-Member
- Site/Operations Medical Officer-Member
- Crew Boat Driver/Quartermaster-Member

COMPANY site representative/Operations Team Leader will be in full command + control during emergencies.

All members of the rescue team must have attended company first aid and firefighting courses and must be certified swimmers.

The following procedure shall apply in an emergency:

- Free any trapped personnel.
- Evacuate the injured to the site clinic (during construction) or to a safe location where the injured can be administered first aid.
- Staff nurse or trained first aiders shall attend to the injured pending transporting to the base clinic at the Methanol Complex in Akabel.
- Arrest the situation by all available means i.e. fire extinguishers, etc.
- Muster personnel at a pre-determined point.
- Take the roll call.
- If necessary, dispatch a search party for any missing personnel.
- Prevent entry by any unauthorised persons.

If the situation threatens personal life, evacuate the scene by means of the crew boat. On stabilization of the affected personnel and departure from site, inform the base officer of the situation and your route of escape if possible.

3.2. Medical Emergency/ Evacuation

The following step shall be taken in the event of an accident requiring evacuation of victims.

Evacuation from site is only through water transportation, after first aid has been administered by the site nurse to arrest bleeding and reduce pain.

- The victim must be properly and adequately bandaged according to the ethics of the nursing profession.
- Movement of the victim to the standby boat shall be by means of a stretcher if available.

- A trained first aider or the site nurse must accompany the Victim.
- If the site nurse accompanies the victim, a nominated first aider shall hold brief for him pending his arrival.
- The base transport officer and staff nurse must be contacted immediately the victim leaves the site.
- The transport officer shall prepare the ambulance ready to receive the victim at the jetty and convey the victim to the company's clinic.
- The base staff nurse shall accompany the victim to the clinic.
- The company's base doctor takes over on receiving the victim.

3.3. Sinking Houseboat (During Construction)

When a houseboat is sinking, the following procedure shall be adopted by the rescue team.

- Raise alarm
- Muster the entire workforce at a pre-determined point
- Leader of rescue team to allocate workforce to available lifesaving floating devices taking care not to exceed the capacity
- Launch lifesaving apparatus with the leader of each group holding the line.
- Workforce, kited with the jackets then enter the life buoys or rafts allocated.
- Meanwhile, other members of the rescue team are to save property until the houseboat is evacuated or it becomes unsafe to continue the recovery procedure.
- The rescue team is the last to leave the houseboat. They should be fully kited and provided with the rescue team life raft.
- All workforce are to be certified swimmers.
- Floodlights should be directed to the area of ultimate safety ashore.

3.4. Fire Emergency

- Sound the alarm
- Muster the workforce at a pre-determined muster point.
- Call for outside help.
- Fight the fire with the available means, while awaiting help from outside. If the fire gets out of hand, the team should leave the vicinity of the accident for their own safety.

- Evacuate the mustered workforce from the muster point to a point of ultimate safety by means of the crew speedboat.
- In case of any injury, send the victim to a safe place for first aid treatment by the nurse. Severe cases should be sent immediately to the base clinic at the methanol Complex in Akabel.
- After the fire is out, salvaging and clearing of debris should be immediately tackled.
- Arrange alternative accommodation and resume normal operation (during construction).

3.5. Snake Bite

Administer first aid as follows:

- Tie firmly the part of the body immediately above the point of bite.
- Incise the point of bite with a sharp razor blade in a 'zig-zag' manner to allow blood to flow out.
- Rush the victim to the clinic as soon as possible.

3.6. Drowning

Any personnel observed drowning, the following procedure shall apply:

- Raise alarm by shouting "man-over board".
- Launch a life-buoy while holding firmly to the securing line and send for the site nurse/trained first aider.
- With assistance from other personnel, carry the victim ashore.
- The site nurse together with trained first aiders is to administer first aid.
- Arrange transport and send the victim to the base clinic in Akabel for further medical attention.

3.7. Speedboat Accident

The general transport officer is to keep a record of the approximate time of departure and arrival of all crafts. On departure of any boat, a message is sent to the location stating the estimated time of arrival. On arrival on site, another message is sent back to the transport department reporting its safe arrival. In case of any undue delay by any water transport vessel in communicating with the base station as at when due, an accident is immediately suspected, a search-team is dispatched, and their findings reported back.

A speedboat accident is considered an emergency and the following steps shall be taken:

- At the scene of accident, the un-injured is to attend to the injured personnel while awaiting help from outside.
- Rescuers (who also form part of the search team) are dispatched to the scene of the accident.
- Victims are immediately transported to the base clinic for further attention.
- Members of the rescue team (including a boat driver) are to stay behind to salvage as well as recover tools and property.

3.8. Epidemy

- Isolate the first victim
- Send the first victim immediately to the base clinic.
- Inoculate all other personnel against the disease or infection
- Eradicate the source of epidemic.

3.9. Armed Robbery Attack on Site

The ultimate objective is to save your life and avoid being harmed, so the following procedure shall apply:

- Do not cause any alarm while the bandits are in operation.
- Obey all their instructions and/or commands.
- Do not try to pull any fast tricks
- After their departure, raise the alarm and inform the base officer through radio communication.
- Check for injuries to personnel. If any send to the site nurse/trained first aider for first aid.
- Severe cases should be sent immediately to the base clinic.
- Inform the law enforcement agents as well as the company security forces.
- Amend damage and return to normal operation.

3.10. Environmental Emergency

An environmental emergency occurs when a harmful substance, such as hydrocarbon liquid or chemical, spills into the environment.

The following steps will be taken in the event of a spill:

- Minimize the area of contamination by containing the spill. Under no circumstances should

the safety of the personnel be jeopardized in securing the area.

- Start spill clean-up using available spill clean-up kit. Call for additional kit if required.
- Where possible, take photographs of the spill. The photographs may assist the spill investigation team.
- Assess the quantity of product spilled and damage done to the environment and document it.
- The Site Team Leader will inform the COMPANY Representative who will in turn notify the relevant Nigerian Regulatory bodies.

4. Disaster Management

Disaster management is the coordination and integration of all activities necessary to build, sustain, and improve the capability to prepare for, protect against, respond to and recover from threatening or actual natural or human-induced disasters.

There are five (5) phases of disaster management, namely:

4.1. Prevention

Prevention focuses on preventing hazards from occurring, whether they are natural, technological or caused by humans. Not all hazards are preventable, but the risk of loss of life and injury can be limited with good evacuation plans, environmental planning and design standards.

4.2. Mitigation

Mitigation is the effort to reduce loss of life and property by lessening the impact of disasters and emergencies. It refers to measures or actions that can prevent an emergency, reduce the chance of an emergency or reduce the damaging effects of unavoidable emergencies. The establishment of building codes and zoning requirements or the creation of defensible space around facilities to protect them from wildfire are examples of mitigation efforts.

4.3. Preparedness

Preparedness is a continuous cycle of planning, organizing, training, equipping, exercising, evaluating and taking corrective action. Training and exercising plans is the cornerstone of preparedness, which focuses on readiness to respond to all hazards, incidents and emergencies. Training and emergency preparedness plans increase a community's ability to respond when a disaster occurs.

Typical preparedness measures include developing mutual aid agreements and memorandums of understanding, training for both response personnel and concerned citizens, conducting disaster exercises to reinforce training and test capabilities, and presenting all-hazards education campaigns.

4.4. Response

The response phase is a reaction to the occurrence of a catastrophic disaster or emergency. It consists of actions which are aimed at saving lives, reducing economic losses and alleviating suffering. The response phase comprises the coordination and management of resources utilizing the Incident Command System. Response actions may include activating the emergency operations center, evacuating threatened populations, opening shelters and providing mass care, emergency rescue and medical care, firefighting, and urban search and rescue.

4.5. Recovery

Recovery consists of those activities that continue beyond the emergency period to restore critical community functions and begin to manage stabilization efforts. The recovery phase begins immediately after the threat to human life has subsided. The goal of the recovery phase is to bring the affected area back to some degree of normalcy, including the restoration of basic services and the repair of physical, social and economic damages. Typical recovery actions include debris cleanup, financial assistance to individuals and governments, rebuilding of roads and bridges and key facilities, and sustained mass care for displaced human and animal populations.

4.6. Disaster Management Plan

Disasters usually occur on a wide scale and cover vast areas hence disaster management is not handled at project or facility level but at Company Corporate level, or higher (Local Government, State Government and Federal Government).

A Disaster Management Plan incorporating the five phases of disaster management highlighted above and taking into consideration the Company's disaster management philosophy and/or policy, will have to be prepared by Brass Fertilizer and Petrochemical Company Limited at corporate level.

The corporate disaster management plan will be used to manage disasters in its areas of operation and/or interest, in conjunction with the relevant local, state and national authorities.